# Explosive Collisions: Momentum Conservation in Reverse

Momentum & Energy: Explosive Collisions · oPhysics

| Senior Secondary (Years 11-12) | ~30 minutes | Science · Physics |
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This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://ophysics.com/e2a.html
Curriculum links: momentum

## Learning intentions

- I can apply conservation of momentum to a system that starts at rest and separates into two moving parts.
- I can explain why kinetic energy increases in an explosive separation.
- I can predict the ratio of the two resulting speeds from the ratio of the two masses.

## Getting started

Open the simulation with two masses initially at rest and touching, with an explosive charge between them. Set the two masses first, then adjust the explosive energy and run the simulation.

## Tasks

**1.** Set the two masses to be equal and choose a moderate explosive energy. Predict whether the two resulting speeds will be equal, then run the simulation to check.

**2.** Set m1 to be double m2, using the same explosive energy. Predict which mass will move faster and by what ratio, using conservation of momentum, then verify.

*Working:*

**3.** Keeping the explosive energy constant, try three different mass ratios and record the resulting speed of each mass.

| m1 : m2 ratio | Speed of m1 | Speed of m2 |
| --- | --- | --- |
| | | |
| | | |
| | | |

**4.** Calculate the total kinetic energy of the system before and after the explosion for one of your trials. Where did this extra energy come from?

*Working:*

## Extension (optional)

Relate this simulation to how a rocket accelerates in space by expelling exhaust gas, even though there is nothing outside the rocket-exhaust system to push against.